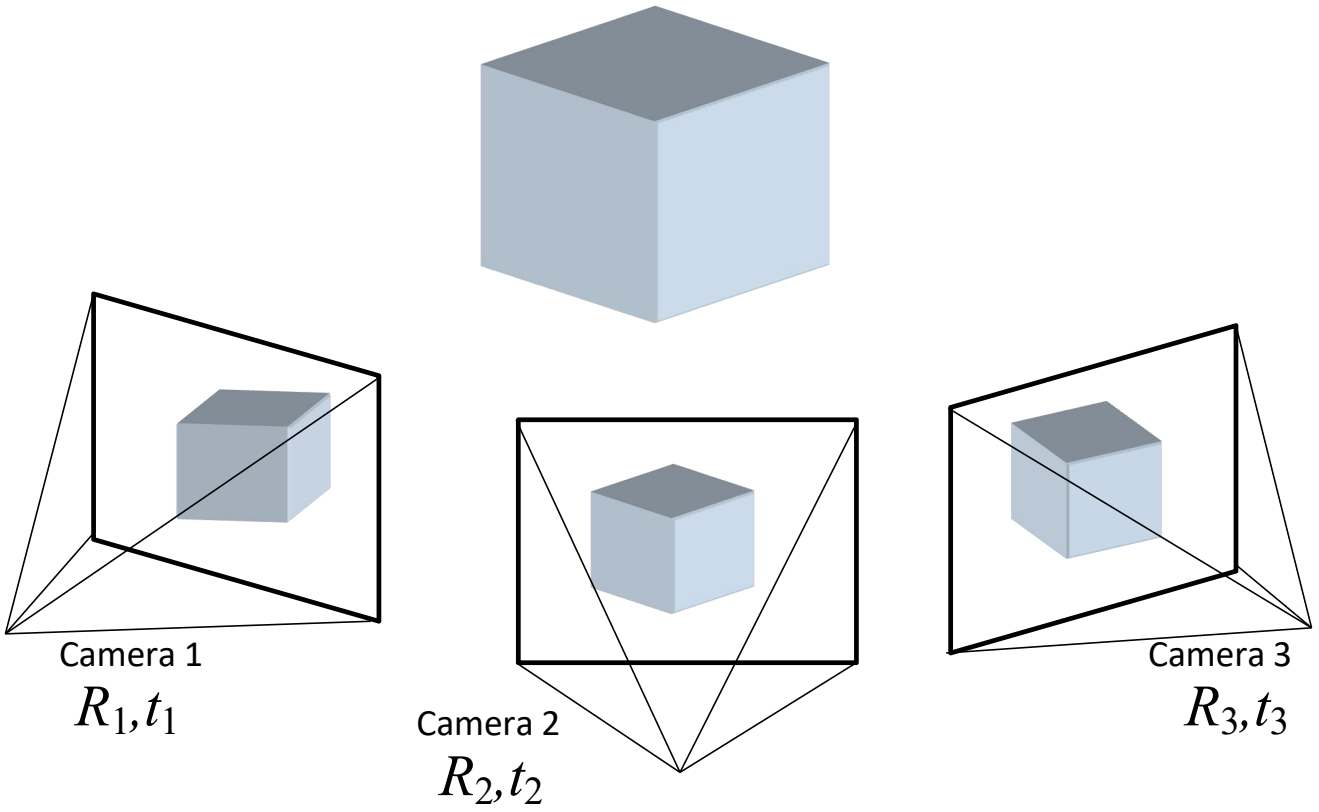
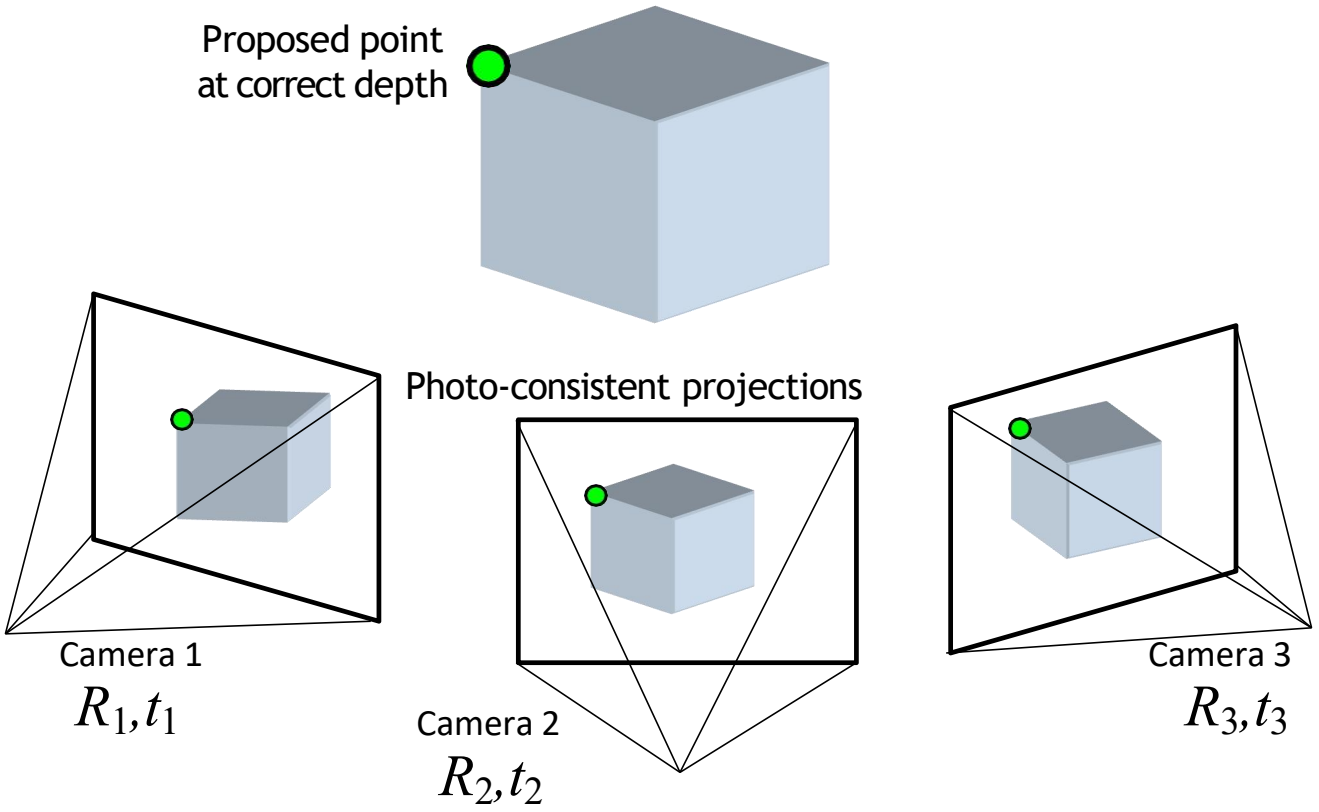
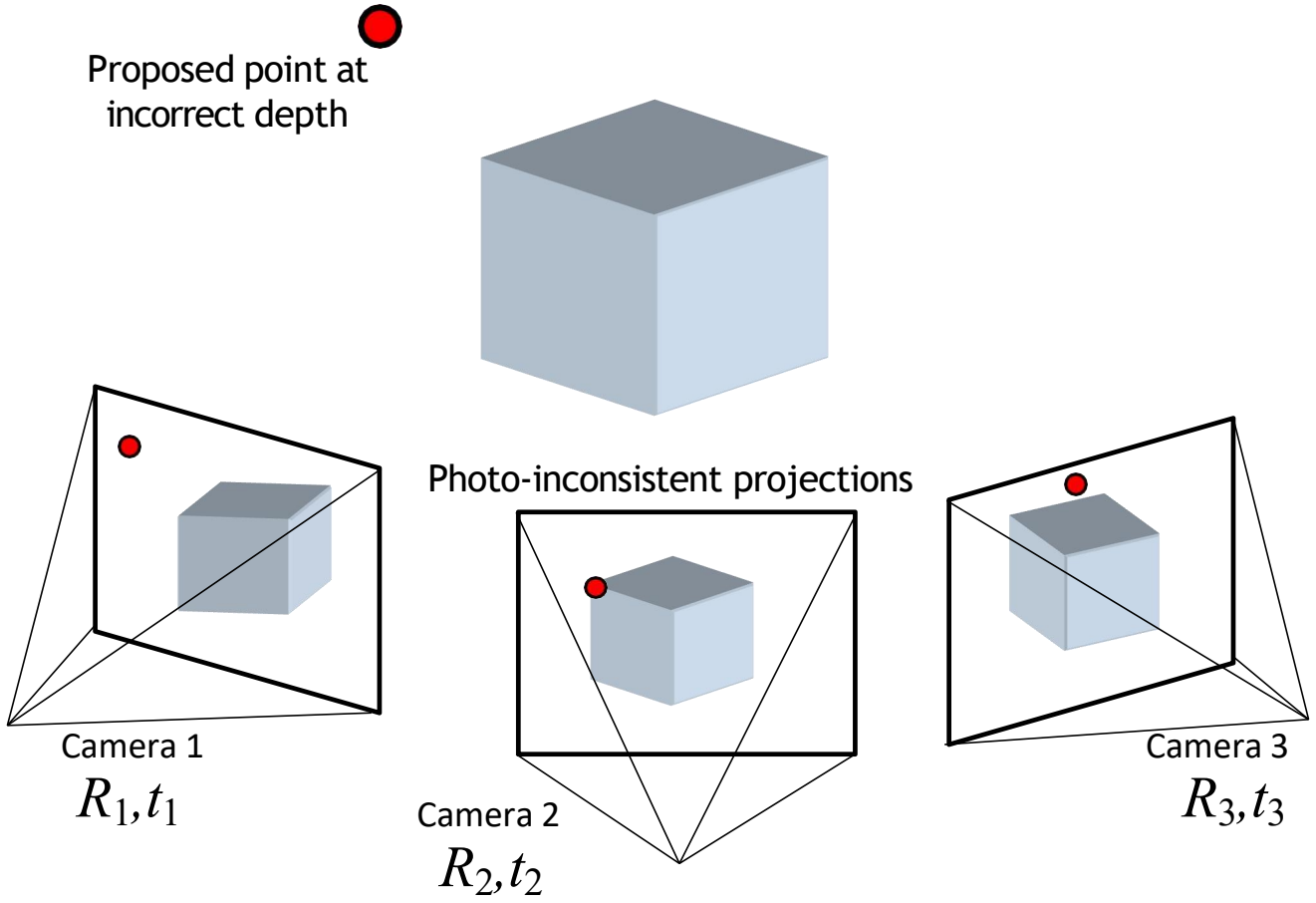
Plane-Sweep Stereo



Plane-Sweep Stereo

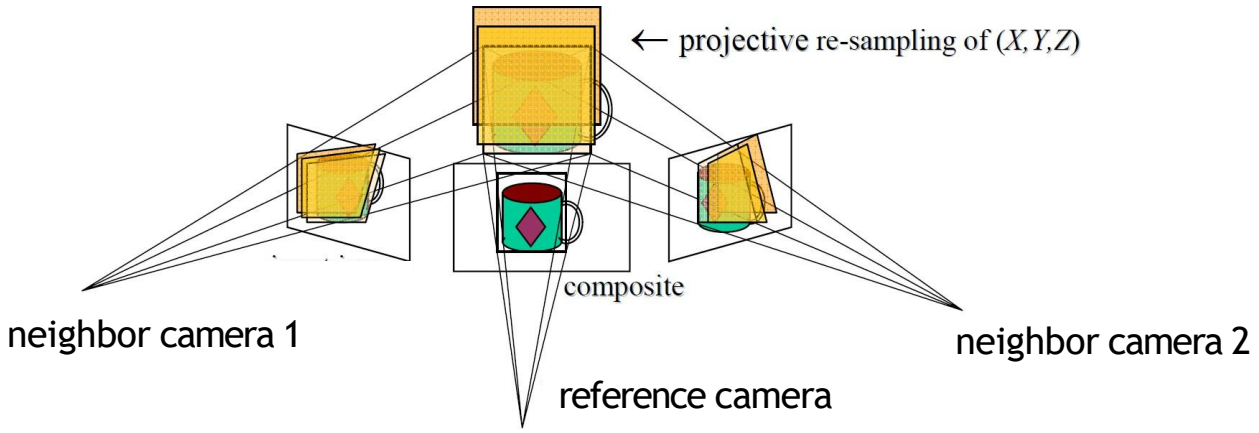


Plane-Sweep Stereo



Plane-Sweep Stereo

- Sweep family of planes parallel to the reference camera image plane
- Reproject neighbors onto each plane (via homography) and compare reprojections



Plane-Sweep Stereo



Left neighbor



Reference image



Right neighbor



Left neighbor projected into reference image

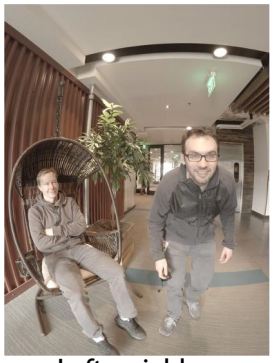


Average images on each plane



Right neighbor projected into reference image

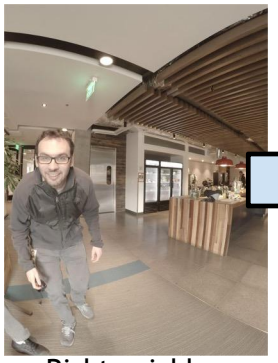
Another example



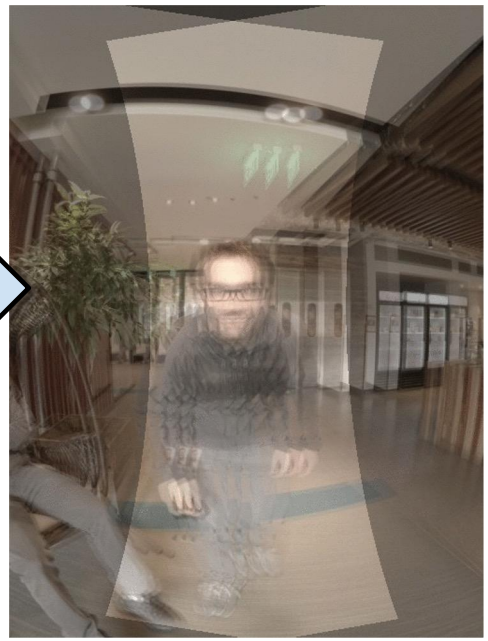
Left neighbor



Reference image

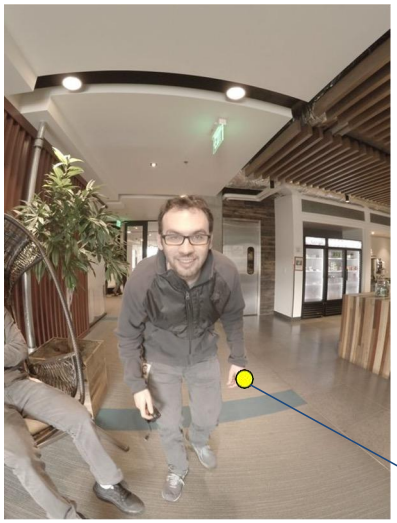


Right neighbor



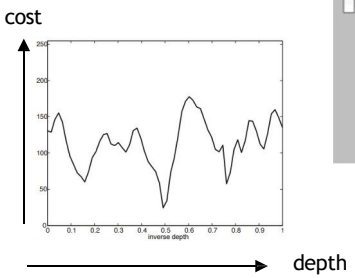
Planar image reprojections swept over depth (averaged)

Cost Volumes -> Depth Maps

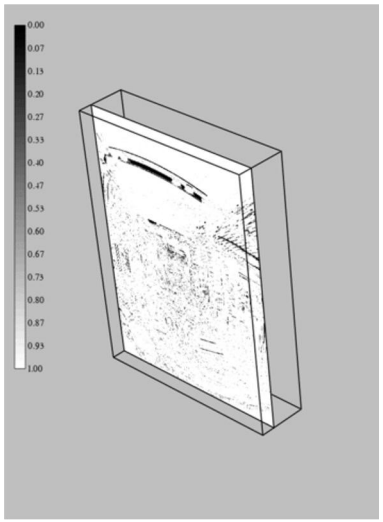


Reference image

Plane sweep



Single pixel's cost profile



Full cost volume

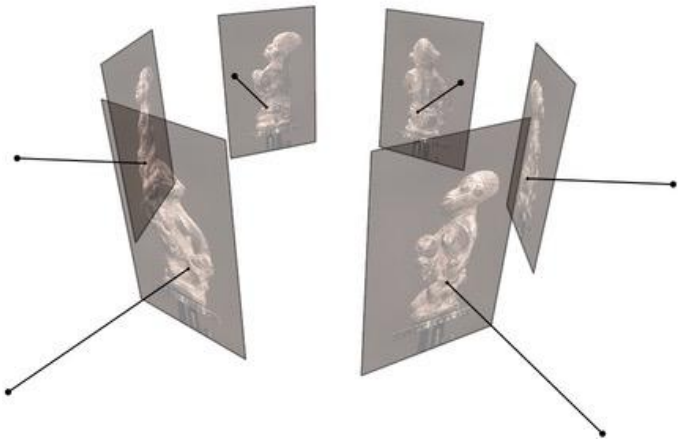
Depth map solver

(Belief propagation, graph cuts, etc.)



Fusing multiple depth maps

- Compute depth map per image
- Fuse the depth maps into a 3D model



Figures by Carlos Hernandez